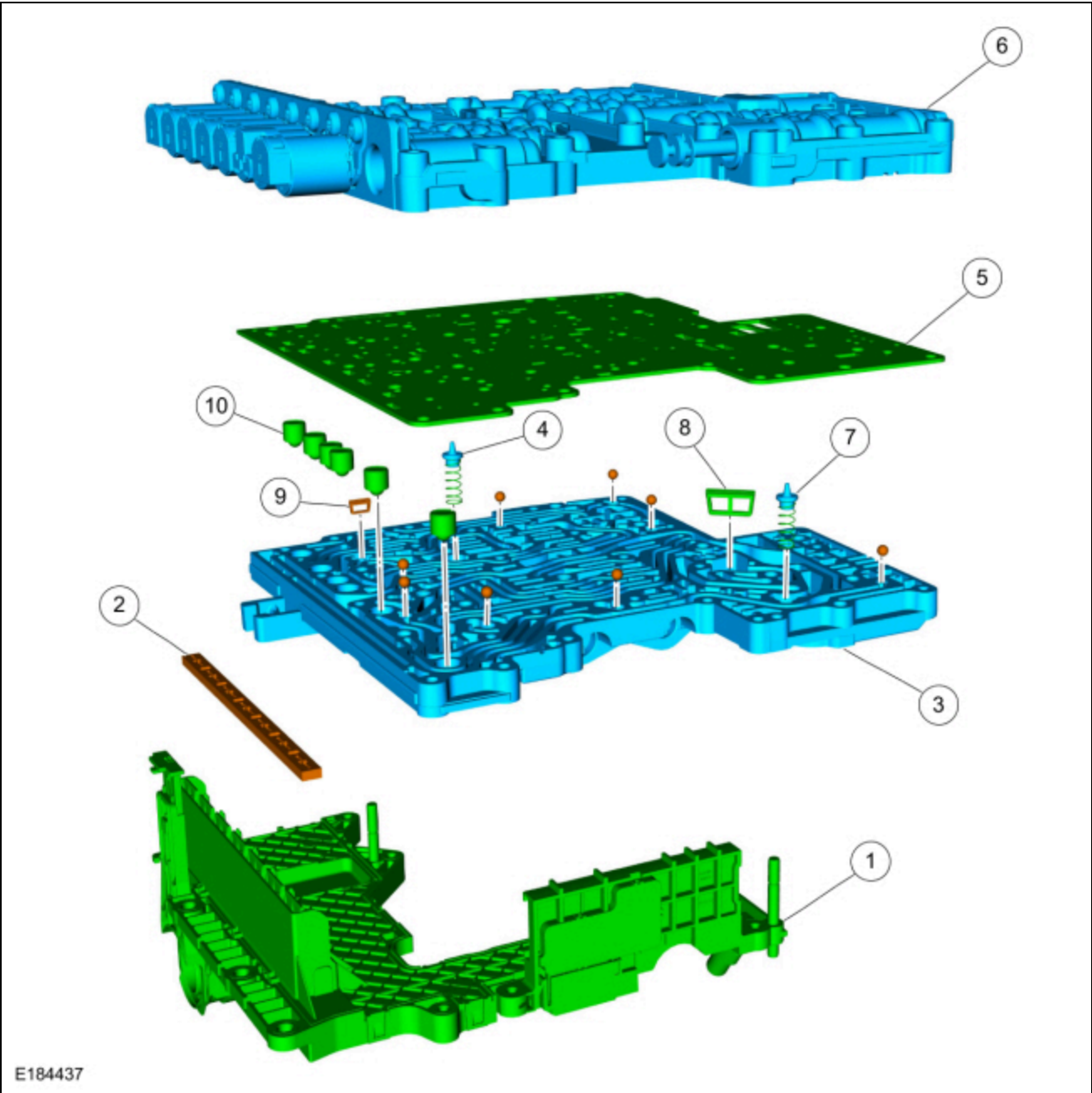


Main Control

 [Ford Ecat Electronic Parts Catalog](#)

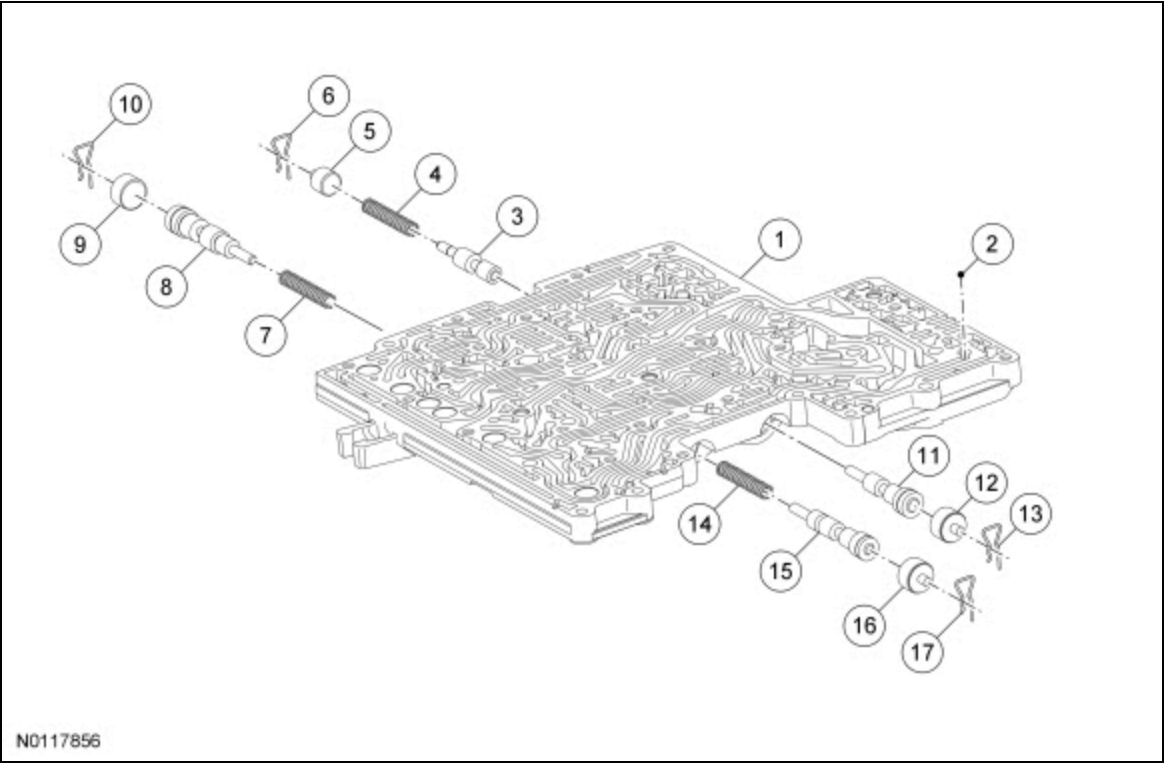
Material	
Item	Specification
Motorcraft® MERCON® LV Automatic Transmission Fluid XT-10-QLVC (US); CXT-10-LV12 (Canada)	MERCON® LV

Main Control Assembly



Item	Part Number	Description
1	7G276 7G276	Molded leadframe
2	—	Foam strip seal
3	—	Main control valve body (upper)
4	7E195 7E195	Main control valve check ball (8 required)
5	7Z490 7Z490	Main control valve body separator plate
6	—	Main control valve body (lower)
7	—	Main control valve
8	7H187 7H187	Main control filter
9	7B155 7B155	Transmission fluid filter
10	7J191 7J191	Transmission solenoid damper valve assembly

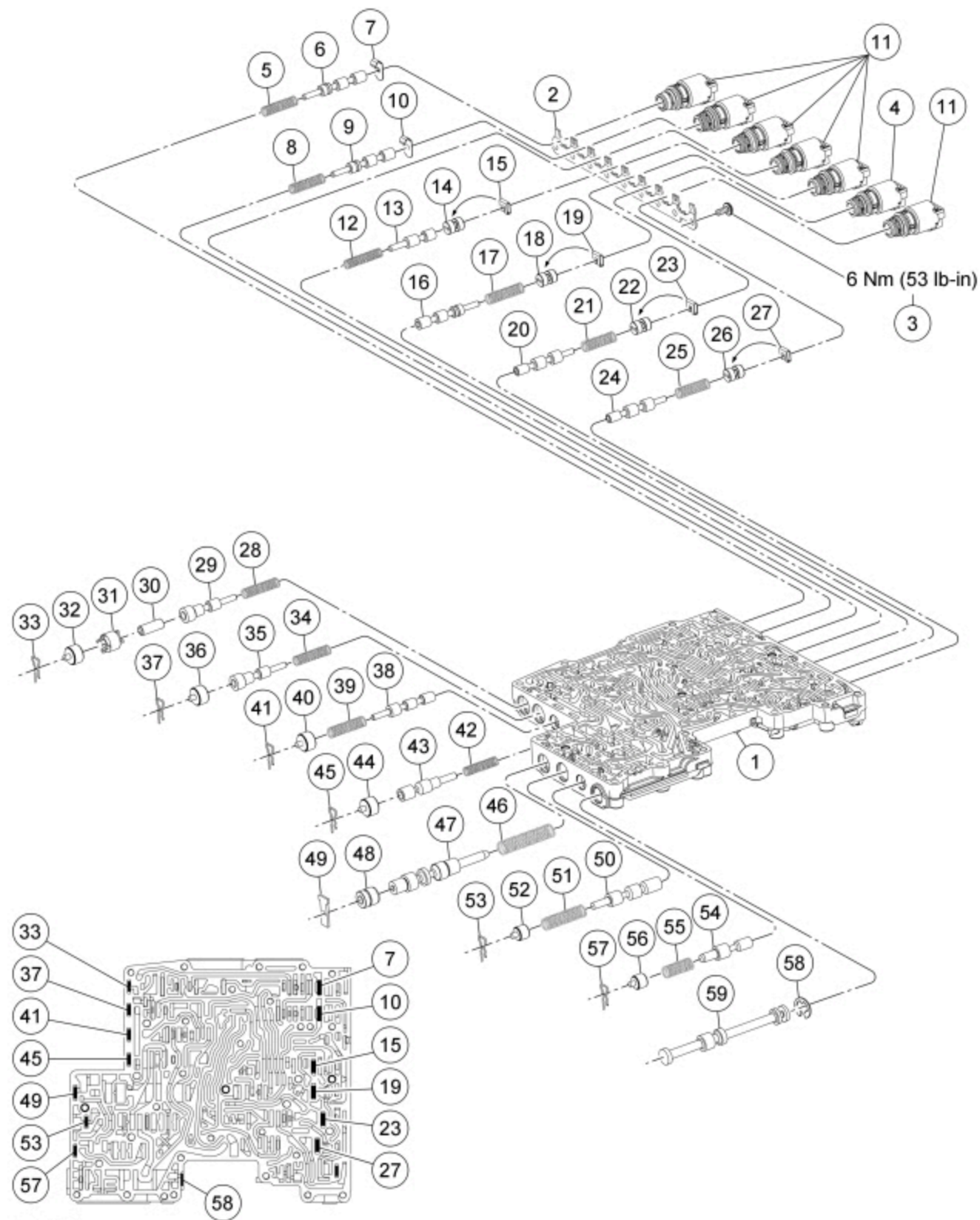
Main Control Upper Half



Item	Part Number	Description
1	7A092 7A092	Main control valve assembly (ditch plate) (upper half)
2	7E195 7E195	Main control valve check ball (8 required)
3	—	Low/reverse (D2) clutch latch valve
4	—	Low/reverse (D2) clutch latch valve spring
5	—	Low/reverse (D2) clutch latch valve plug
6	7G007 7G007	Low/reverse (D2) clutch latch valve plug retaining clip
7	—	Direct (B) clutch regulator valve spring
8	—	Direct (B) clutch regulator valve
9	—	Direct (B) clutch regulator valve plug

10	7G007 7G007	Direct (B) clutch regulator valve plug retaining clip
11	—	Low/reverse (D1) clutch regulator valve
12	—	Low/reverse (D1) clutch regulator valve plug
13	7G007 7G007	Low/reverse (D1) clutch regulator valve plug retaining clip
14	—	Intermediate (C) clutch regulator valve spring
15	—	Intermediate (C) clutch regulator valve
16	—	Intermediate (C) clutch regulator valve plug
17	7G007 7G007	Intermediate (C) clutch regulator valve plug retaining clip

Main Control Lower Half



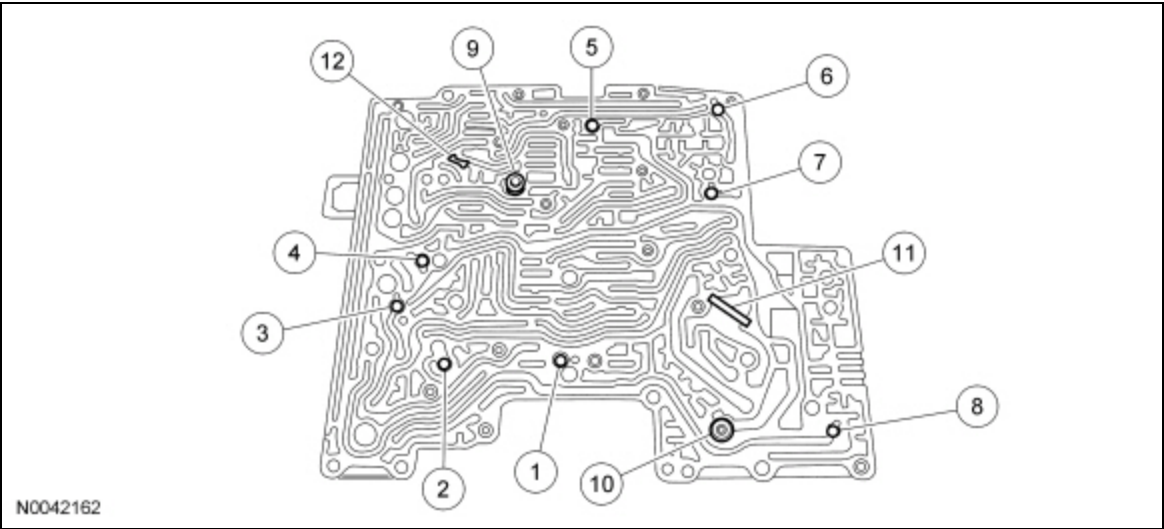
N0117857

Item	Part Number	Description
1	—	Main control valve assembly (lower half)
2	—	Transmission shift control solenoid hold-down bracket
3	—	Transmission shift control solenoid hold-down bracket bolt M5 x 0.80 x 12 (8 required)
4	7G4847G484	Transmission ON/OFF shift solenoid
5	—	Forward (A) clutch latch valve spring
6	—	Forward (A) clutch latch valve
7	—	Forward (A) clutch latch valve retainer plate

8	—	Direct (B) clutch latch valve spring
9	—	Direct (B) clutch latch valve
10	7F194 7F194	Direct (B) clutch latch valve retainer plate
11	7G383 7G383	Transmission VES (6 required)
12	—	Solenoid regulator valve spring
13	—	Solenoid regulator valve
14	—	Solenoid regulator valve plug
15	7G007 7G007	Solenoid regulator valve plug retainer
16	—	Low/reverse (D1) clutch latch valve
17	—	Low/reverse (D1) clutch latch valve spring
18	—	Low/reverse (D1) clutch latch valve plug
19	7G007 7G007	Low/reverse (D1) clutch latch valve plug retainer
20	—	Drive enable valve
21	—	Drive enable valve spring
22	—	Drive enable valve plug
23	7G007 7G007	Drive enable valve plug retainer
24	—	Solenoid multiplex valve
25	—	Solenoid multiplex valve spring
26	—	Solenoid multiplex valve plug
27	7G007 7G007	Solenoid multiplex valve plug retainer
28	—	Forward (A) clutch regulator valve spring
29	—	Forward (A) clutch regulator valve
30	—	Forward (A) clutch regulator valve sleeve
31	—	Forward (A) clutch regulator valve return plug
32	—	Forward (A) clutch regulator valve plug
33	7G077 7G077	Forward (A) clutch regulator valve plug retaining clip
34	—	Overdrive (E) clutch regulator valve spring
35	—	Overdrive (E) clutch regulator valve
36	—	Overdrive (E) clutch regulator valve plug
37	7G007 7G007	Overdrive (E) clutch regulator valve plug retaining clip
38	—	Overdrive (E) clutch latch valve
39	—	Overdrive (E) clutch latch valve spring
40	—	Overdrive (E) clutch latch valve plug
41	7G007 7G007	Overdrive (E) clutch latch valve plug retaining clip
42	—	Bypass clutch control regulator valve spring
43	—	Bypass clutch control regulator valve
44	—	Bypass clutch control regulator valve plug
45	7G007 7G007	Bypass clutch control regulator valve plug retaining clip
46	—	Main regulator valve spring

47	—	Main regulator valve
48	—	Main regulator valve sleeve
49	7G0077G007	Main regulator valve clip
50	—	Converter release regulator valve
51	—	Converter release regulator valve spring
52	—	Converter release regulator valve plug
53	7G0077G007	Converter release regulator valve plug retaining clip
54	—	Lubrication control valve
55	—	Lubrication control valve spring
56	—	Lubrication control valve plug
57	7G0077G007	Lubrication control valve plug retaining clip
58	W527007W527007-S-S	Retaining ring
59	—	Transmission control manual valve

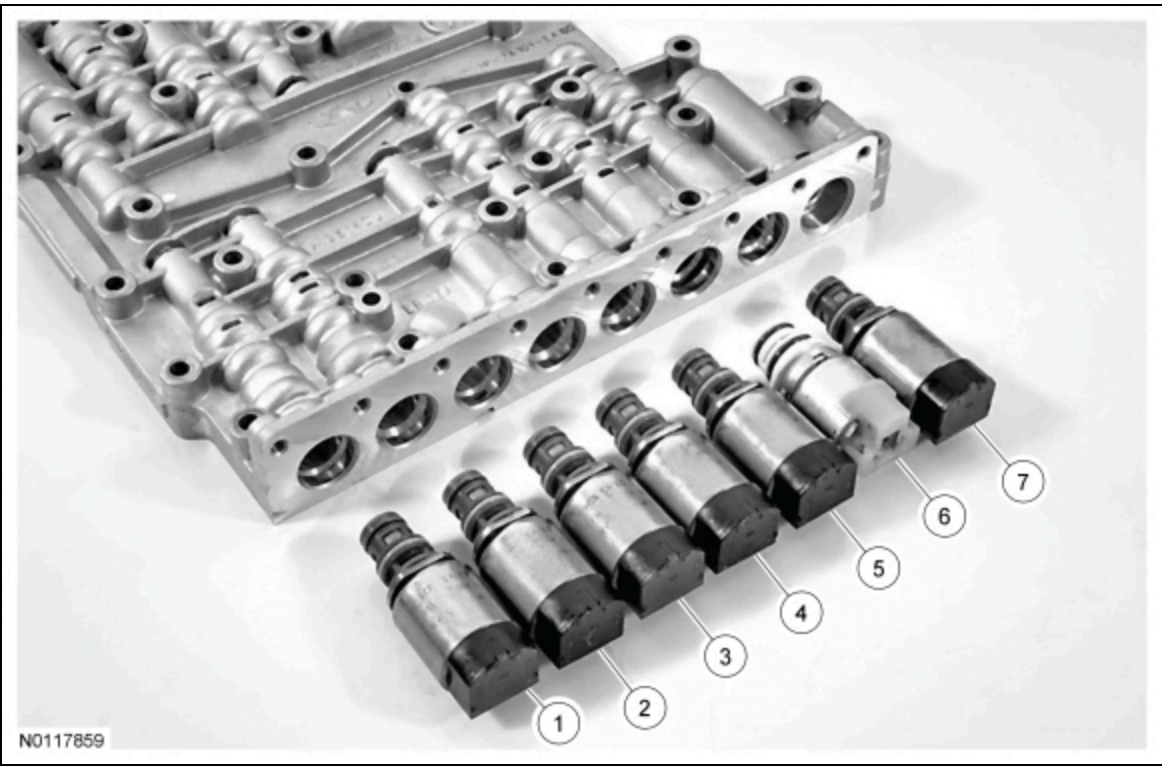
Check Ball Location



Item	Description
1	Reverse/drive control check ball
2	Shift solenoid control check ball
3	Intermediate clutch exhaust control check ball
4	Direct clutch exhaust control check ball
5	Low/reverse clutch exhaust control check ball
6	Forward clutch exhaust control check ball
7	Overdrive clutch exhaust control check ball
8	Reverse exhaust control check ball
9	Clutch exhaust valve
10	Converter drain back valve

Item	Description
11	Large filter screen
12	Small filter screen

Shift Solenoid Location



Item	Part Number	Description
1	7G3837G383	SSA , Forward (1,2,3,4) clutch, normally low VFS (brown nozzle).
2	7G3837G383	SSB , Direct (3,5,R) clutch, normally high VFS (black nozzle).
3	7G3837G383	SSC , Intermediate (2,6) clutch, normally low VFS (brown nozzle).
4	7G3837G383	LPC solenoid, normally high VFS (black nozzle).
5	7G3837G383	SSD , low/reverse/overdrive clutch, normally high VFS (black nozzle).
6	7G4847G484	SSE , normally off ON/OFF solenoid.
7	7G3837G383	TCC solenoid, normally low VFS (brown nozzle).

Disassembly

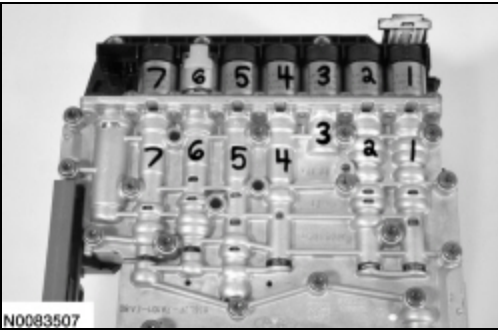
NOTE: Make an identifying mark on each solenoid and the corresponding bore for correct assembly.

NOTE: Solenoids may visually appear the same but their designs/functions are different. Use care not to assemble the main control assembly incorrectly. Incorrect solenoid installation results in poor transmission shift quality.

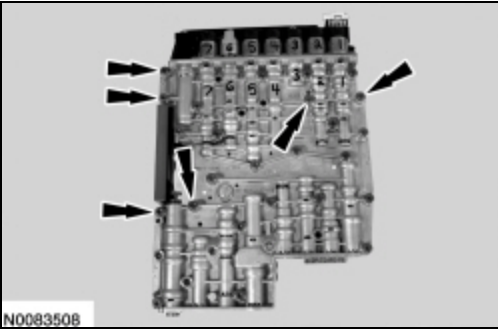
NOTE: Note the location of the 8 check balls, 6 solenoid dampers, 2 internal valves and springs and 2 filter screens for reassembly.

Main Control Upper Half

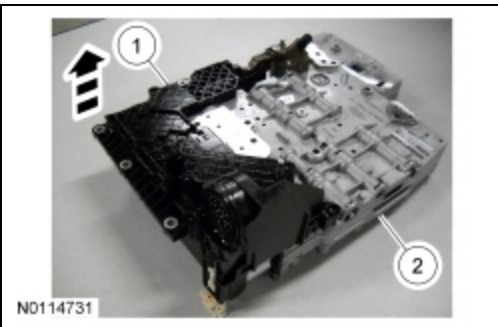
1. Make an identifying mark on each solenoid and the corresponding bore for correct assembly.



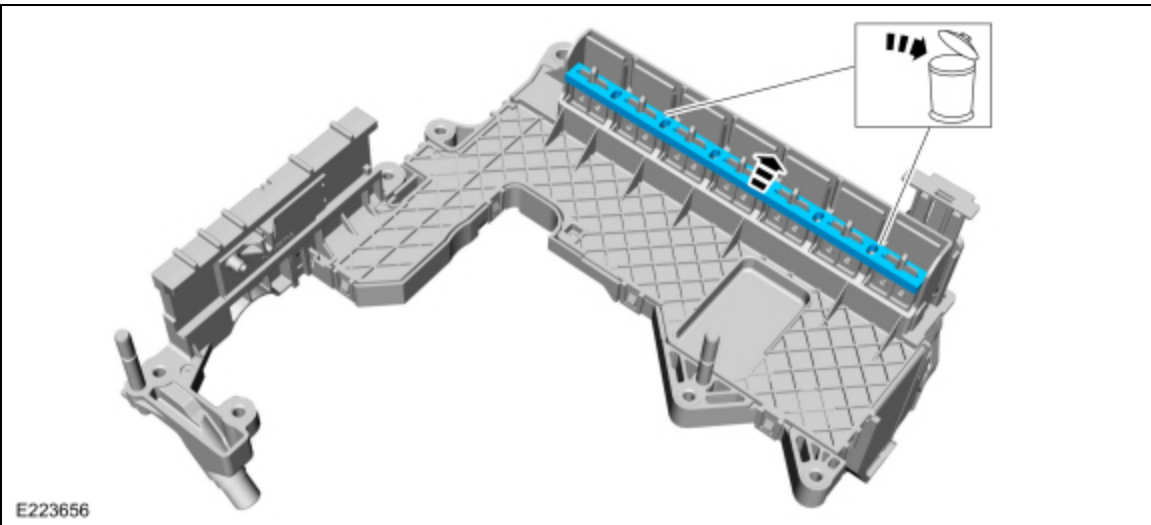
2. Remove the 6 long molded leadframe-to-valve body bolts from the molded leadframe.



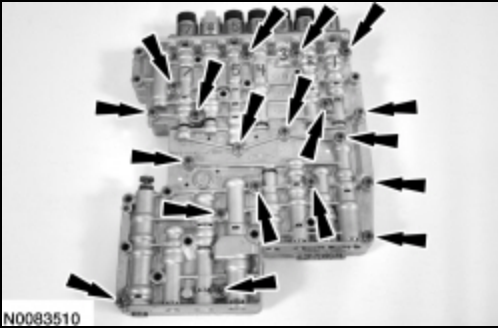
3. Carefully separate the molded leadframe from the main control assembly.
 1. Molded leadframe
 2. Main control assembly



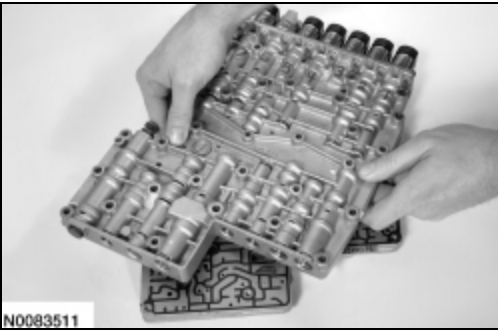
4. If equipped, remove and discard the foam strip seal.



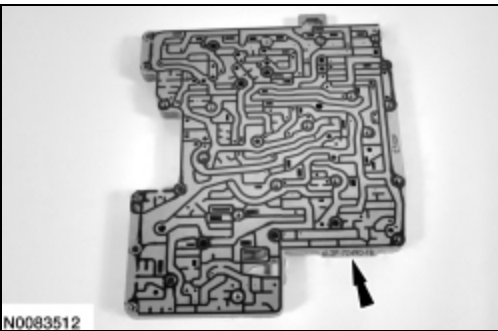
5. Remove 19 short molded leadframe-to-valve body bolts from the main control assembly.



6. Separate the lower half of the main control assembly from the upper half of the main control assembly.

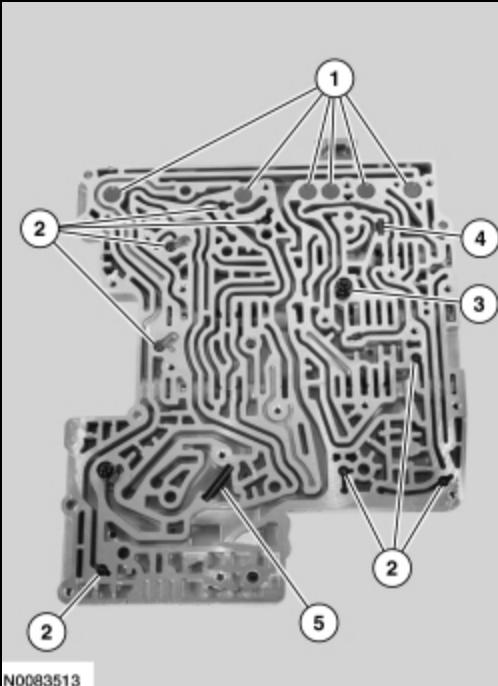


7. Remove and discard the main control valve body separator plate.

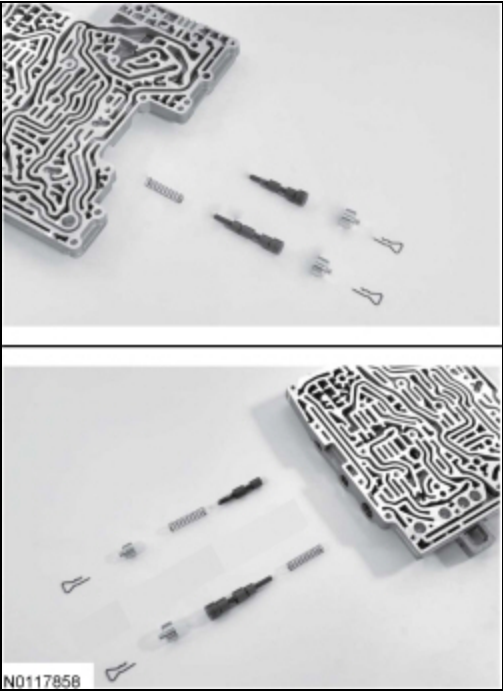


8. Remove the solenoid dampers, check balls, converter drain back valve and filter screens from the main control assembly. Note the location for assembly.

1. Solenoid dampers
2. Check balls
3. Converter drain back valve
4. Small filter screen
5. Large filter screen



9. Remove the retaining clips, caps, valves and valve springs from each bore of the main control assembly. Note the location and order for assembly.



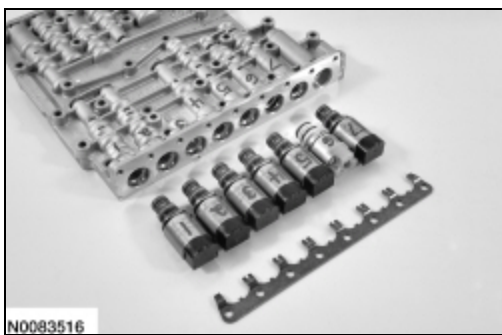
10. Using clean mineral spirits, clean the upper half of the main control assembly.

Main Control Lower Half

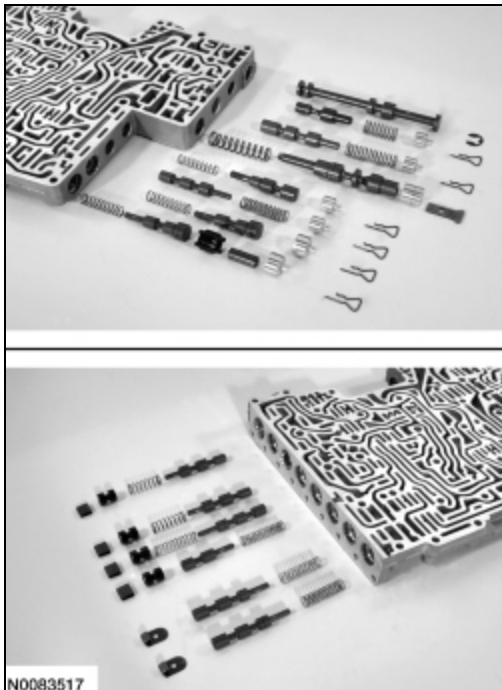
11. Remove the 8 solenoid hold down bracket bolts and the solenoid hold down bracket.



12. Remove the solenoids.



13. Remove the retaining clips, caps, valves and valve springs from each bore of the main control assembly. Note the location and order for assembly.



14. Using clean mineral spirits, clean the lower half of the main control assembly.

Assembly

Main Control Upper Half

1. **NOTE:** *Many components and surfaces in the main control valve body are precision machined. Careful handling during disassembly, cleaning, inspection and assembly prevents unnecessary damage to machined surfaces.*

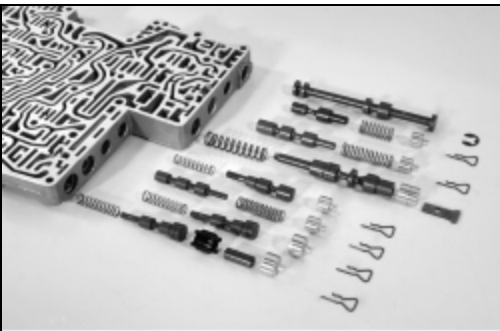
Install the retaining clips, caps, valves and valve spring into each bore.



Main Control Lower Half

2. **NOTE:** *Many components and surfaces in the main control valve body are precision machined. Careful handling during disassembly, cleaning, inspection and assembly prevents unnecessary damage to machined surfaces.*

Install the retaining clips, caps, valves and valve spring into each bore.



3. **NOTICE:** **A .YES is calibrated from the factory and are not all the same. To replace a .YES , match the replacement solenoid nozzle color and the band number with the original solenoid or harsh/erratic shifts, harsh/soft engagement or damage to the transmission can occur.**

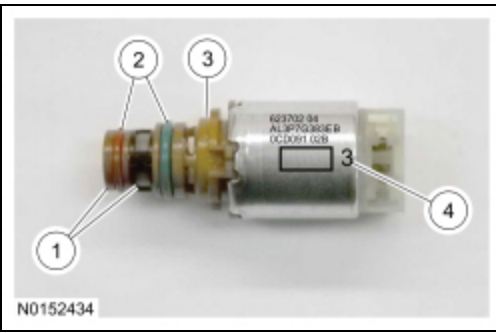
Inspect the solenoid screens for contamination and install new solenoids if needed. Lubricate the solenoid O-rings with clean transmission fluid.

If replacing a solenoid, use the solenoid exploded view at the beginning of this procedure to identify what type of solenoid is being replaced. The solenoid can be a normally high .YES with a black nozzle, normally low .YES with a brown nozzle or an ON/OFF solenoid.

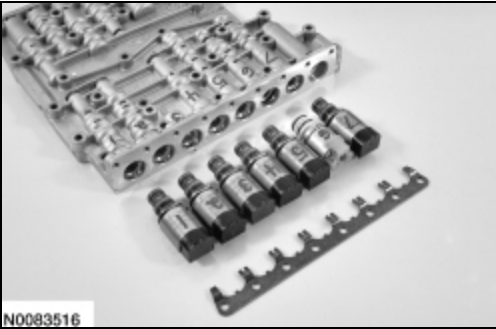
If replacing a .YES , record the band number and the solenoid nozzle color to get the correct part number for the solenoid.

1. Screens

2. O-rings
3. Solenoid nozzle
4. Solenoid band number



4. Install the solenoids into their correct bores as indicated during disassembly.



5. Install the shift control solenoid hold down bracket and the 8 hold down bracket bolts.

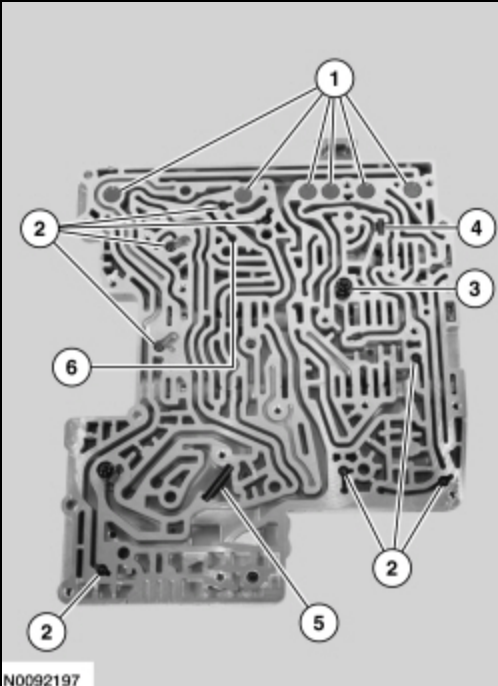
- Tighten to 6 Nm (53 lb-in).



6. NOTICE: Toward the back of the upper valve body there is a hydraulic passage that is shaped as though it receives a check ball. This passage does not require a check ball. Be sure not to mistake this hydraulic passage for a passage that does require a check ball or damage to the transmission can occur.

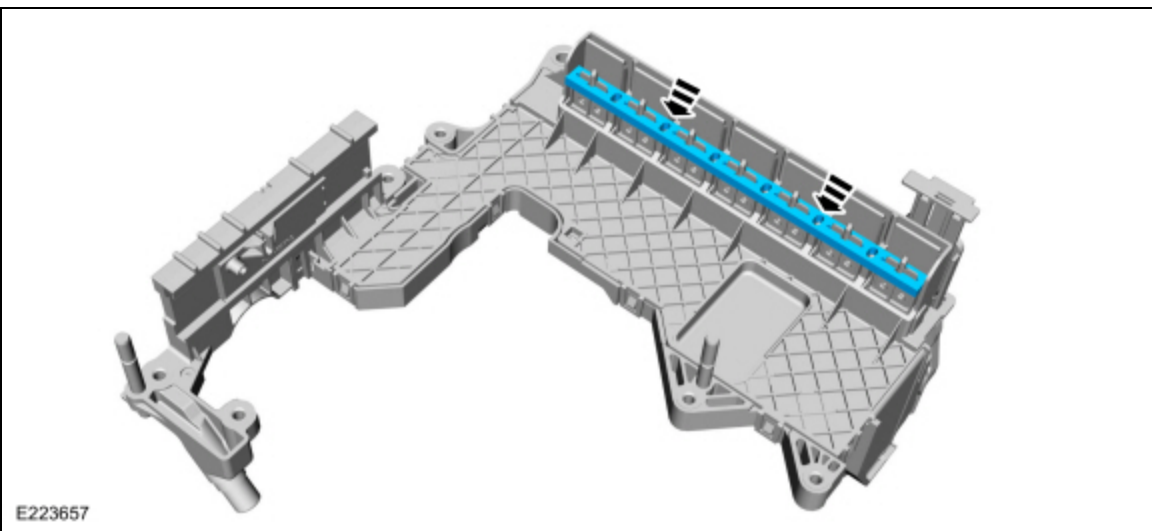
Install the solenoid dampers, check balls, converter drain back valve and filter screens.

1. Solenoid dampers
2. Check balls
3. Converter drain back valve
4. Small filter screen
5. Large filter screen
6. Empty hydraulic passage (no check ball)



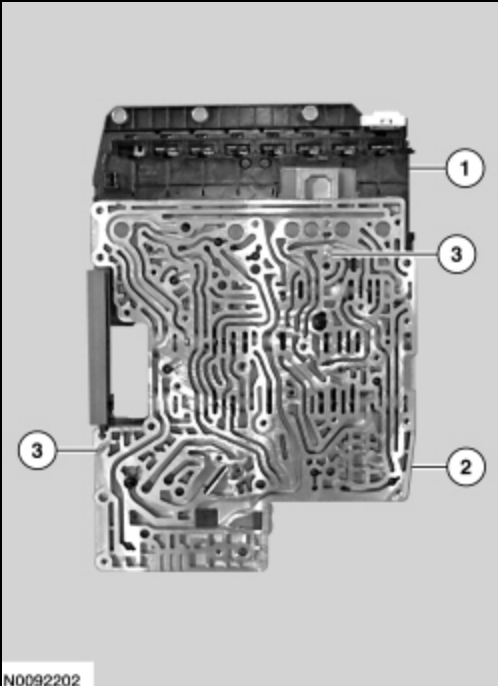
7. **NOTE:** Some vehicles may not have been built with a foam strip seal. A new foam strip seal should be installed on all molded leadframes.

Install a new foam strip seal on the molded leadframe.



8. Align the guide pins on the molded leadframe with the holes in the main control upper valve body and install the valve body onto the molded leadframe.

1. Molded leadframe
2. Main control upper valve body
3. Guide pins

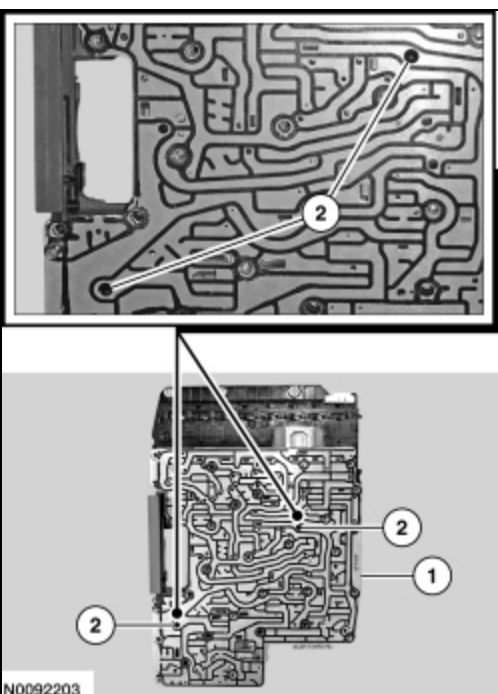


9. **NOTICE:** If the drain back valves are not installed correctly in the separator plate, damage to the transmission can occur.

Position the new main control separator plate on the main control upper valve body.

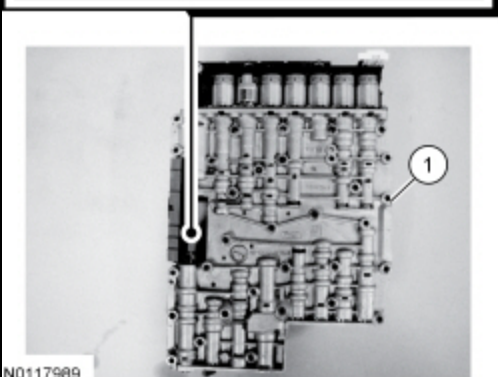
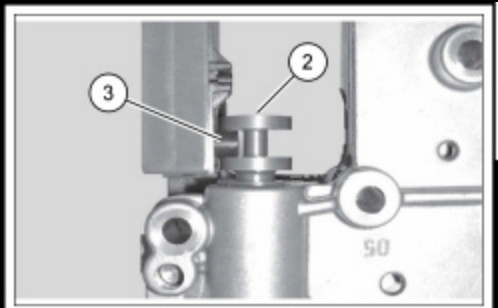
Check the 2 drain back valves to be sure they are correctly positioned in the separator plate when the separator plate is pressed flush against the main control upper valve body.

1. Main control separator plate
2. Drain back valves

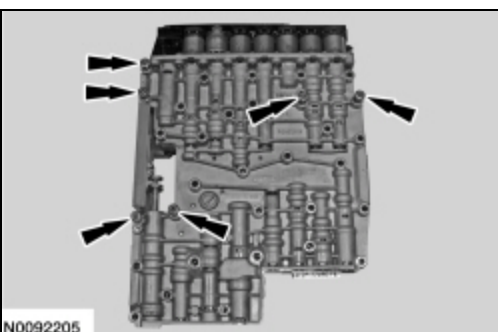


10. Position the main control lower valve body on the 2 molded leadframe alignment studs, align the manual valve with the $\overline{TR}$ sensor on the molded leadframe and install the main control lower valve body.

1. Main control lower valve body
2. Manual valve
3. $\overline{TR}$ sensor



11. Install the 6 long molded leadframe-to-valve body bolts hand-tight.



12. Install the 19 short main control assembly lower-to-upper valve body bolts. Tighten the bolts in the sequence shown.

- Tighten bolts 1 through 6 to 6 Nm (53 lb-in).
- Tighten bolts 7 through 25 to 7 Nm (62 lb-in).

